

Product Requirements Document

Project Name: Mini Interactive Electronic Pet

Group: Session 2 – Group 8

Target Users: Office workers & international students (20–35) with loneliness & emotional companionship needs

1. User Stories

- a. As a lonely office worker / international student, I want the device to react to my touch, shake, tilt and loud sounds with rich expressions, so that I can get instant emotional companionship and fun.
- b. As a daily user, I want to view real-time temperature, humidity, local time and weather on the screen, so that I can get environmental info conveniently while interacting with the pet.
- c. As a user who needs flexible control, I want to long-press Key B to call a QR code, scan it with my phone to remotely control expressions via a web page, so that I can interact without touching the device directly.
- d. As a user who needs emotional communication, I want to press Key A to enter voice interaction mode quickly, so that I can talk to relieve loneliness easily.
- e. As a busy user, I want the device to play automatic idle animations when waiting for interaction, so that it stays lively without any manual operation.
- f. As a non-technical user, I want all interactions to be intuitive and fast-responsive, so that I can use the product without learning burden.

2. Feature List Linked to User Stories

User Story ID	Corresponding Features
1	Multi-sensor expression interaction: tap to smile, shake/flip to dizzy, drag/tilt to follow, loud sound to scare
2	Real-time data display: temperature & humidity (sensor), local time & weather (API)
3	Long-press Key B to generate QR code; mobile scan for web remote expression control
4	One-key (Key A) voice interaction mode (connected to ATOM module)
5	Automatic loop idle animations in standby status
6	Low-latency sensor response, simple operation, no complex setup

3. Initial Technical Architecture

Frontend

- a. Device screen UI: display expressions, animations, temperature, humidity, time, weather
- b. Mobile web page: QR code access, remote control panel

Backend

- a. Sensor data processing (touch, gesture, sound, temperature/humidity)
- b. Expression & animation logic control
- c. ATOM voice module driver
- d. Weather & time API integration

Data Storage

- a. Local lightweight storage: animation frames, interaction parameters
- b. Real-time cache: sensor data, weather, temperature, humidity